

Institute of Actuaries of India

Subject CB2-Business Economics

November 2019 Examination

INDICATIVE SOLUTION

Introduction

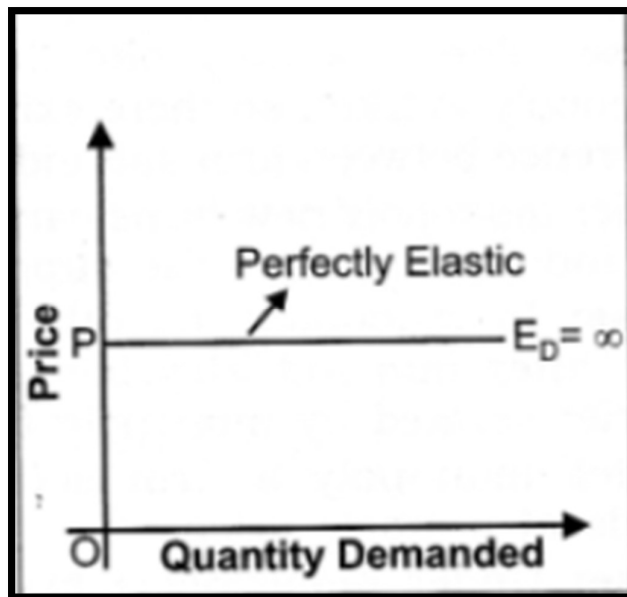
The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

<u>Solution 1:</u>	B	[1.5 Marks]
<u>Solution 2:</u>	B	[1.5 Marks]
<u>Solution 3:</u>	C	[1.5 Marks]
<u>Solution 4:</u>	C	[1.5 Marks]
<u>Solution 5:</u>	C	[1.5 Marks]
<u>Solution 6:</u>	A	[1.5 Marks]
<u>Solution 7:</u>	C	[1.5 Marks]
<u>Solution 8:</u>	A	[1.5 Marks]
<u>Solution 9:</u>	C	[1.5 Marks]
<u>Solution 10:</u>	C	[1.5 Marks]
<u>Solution 11:</u>	B	[1.5 Marks]
<u>Solution 12:</u>	C	[1.5 Marks]
<u>Solution 13:</u>	C	[1.5 Marks]
<u>Solution 14:</u>	C	[1.5 Marks]
<u>Solution 15:</u>	B	[1.5 Marks]
<u>Solution 16:</u>	D	[1.5 Marks]
<u>Solution 17:</u>	C	[1.5 Marks]
<u>Solution 18:</u>	C	[1.5 Marks]
<u>Solution 29:</u>	D	[1.5 Marks]
<u>Solution 20:</u>	D	[1.5 Marks]
<u>Solution 21:</u>	C	[1.5 Marks]
<u>Solution 22:</u>	A	[1.5 Marks]
<u>Solution 23:</u>	C	[1.5 Marks]
<u>Solution 24:</u>	D	[1.5 Marks]
<u>Solution 25:</u>	A	[1.5 Marks]
<u>Solution 26:</u>	A	[1.5 Marks]

Solution 27:

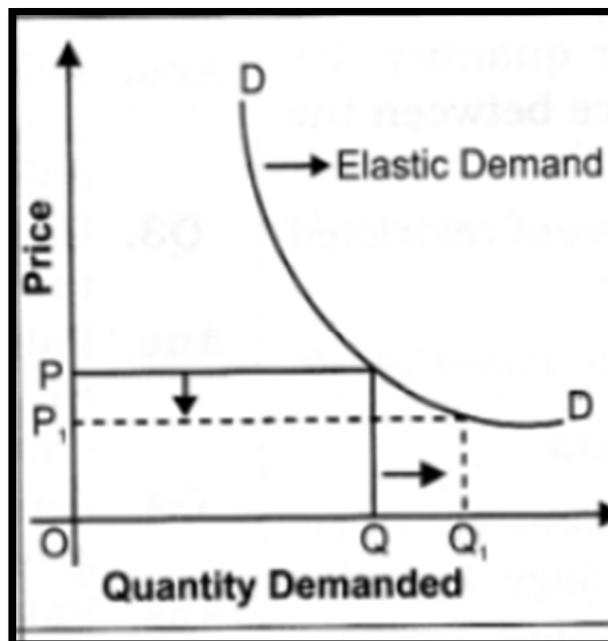
- i) A. Increase in quantity demanded
 B. Increase in supply
 C. Decrease in supply
 D. Decrease in quantity demanded
- ii) a) Under Perfect competition, every firm is a price taker. The price is set by industry demand and supply. The goods produced are homogeneous in nature and price remains constant. Therefore, we have a perfectly elastic demand curve, i.e., $PED = \infty$

[2]



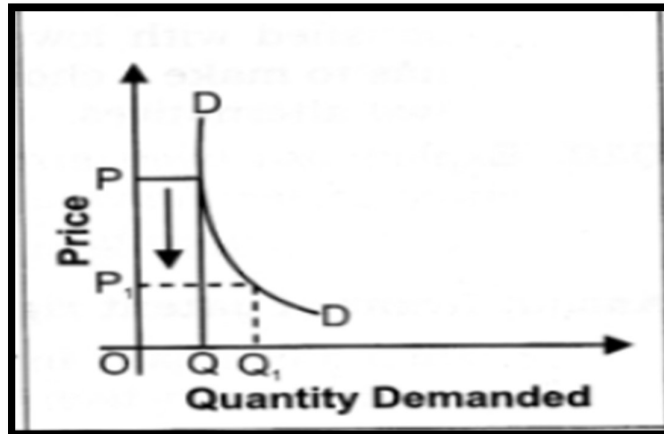
[2]

- b) A monopolistic competitive firm has to design its own pricing strategy. One can expect to sell larger quantity at a lower price and vice versa. Close substitutes are produced, so demand fluctuates more proportionately than the price of the commodity. $PED > 1$



[2]

- c) A monopolist is the only producer of good which has no close substitute. That is why demand fluctuates less proportionately than the price of the commodity. $PED < 1$



[2]

[6]

[8 Marks]

Solution 28:i) **Exceptions to Law of demand**

According to the law of demand, there exists an inverse relationship between price of a commodity and its quantity demanded. However, there are certain exceptions to this rule which are enumerated below.

Conspicuous consumption

Articles of conspicuous consumption or prestigious goods like diamonds, antique pieces etc. are demanded by very rich people for their social prestige. When the price of such goods rise, their use becomes more attractive and they are purchased in larger quantities. It is also known as the Veblen effect, named after Thorsten Veblen (an economist). Consumers under this effect measure the utility of a commodity based on the price. Higher the price, higher is the utility and vice-verse. Thus, consumers buy more of the commodity at a higher price. Diamonds are an example for this. Higher the price of diamonds, higher is the prestige value attached to them and hence higher is the demand for them.

Giffen goods:

Generally, those goods which are considered inferior by the consumers and occupy a substantial place in the consumer's budget are called Giffen goods. Sir Robert Giffen, an economist, found that the people of Britain consumed more bread when the prices went up. This was not as per law of demand. This was because they could not afford to spend on more expensive food items such as meat. Indirectly, it impacted the purchasing power of the consumer. Thus, even though the price of bread went high, it was still cheaper compared to other foods. This forced them to substitute meat and buy more bread. Thus this is a clear exception to the law of demand and hence, the demand curve has a positive slope. When the price of bread decreases, the real income of the consumers would increase and they can afford to buy a more expensive item (here, meat), thus, the demand for bread decreases. Hence such good which exhibit direct price demand relationship are called Giffen goods.

Conspicuous necessities:

Sometimes, demand is affected by the demonstration effect i.e., by imitating the consumption pattern of a social group to which the individual belongs. These goods have become necessities of life due to their constant usage.

Example – Demand for refrigerator, TV sets etc. does not fall even if their price rises. This is because they have become necessities of life due to continuous usage, even though they are luxury goods.

Expectation about future changes in price:

In certain situations, people tend to buy more of a commodity anticipating that in future, the prices will be higher.

Example – In case of shares, if the prices are expected to increase in the future, investors will buy more number of shares now, even if the present price is higher.

Emergency:

In cases of emergency, people will buy the goods no matter how high the prices are.

Example – If a person has to undergo a surgery, he has to buy the medicines required even if their prices are high.

Speculation:

The person will buy more stocks if the price of the security is increasing and wait and does not buy when the price is falling.

Other exceptions

- In ignorance of the ruling price of the commodity, consumers may purchase the commodity even at high prices.
- Consumers sometimes tend to be irrational and make impulsive purchases without calculation of the price and the usefulness of the product.

[4]

ii) Yes, All Inferior goods are necessities but all necessities are not inferior goods.

[1]

If you consume less of a good as income goes up, then it must be true that you spend a smaller fraction of your income on that good as income goes up. Thus, all inferior goods are necessities.

[1]

At the same time, it may be the case that the fraction of your income spent on a good declines as your income goes up — but you still buy more of the good. (For instance, suppose your income goes up by 10% and you choose to consume 5% more of a good. Then the fraction of income spent on that good is declining even though you are increasing your consumption of the good as your income goes up.) Thus, necessities could be normal goods.

[1]

[3]

[7 Marks]

Solution 29:

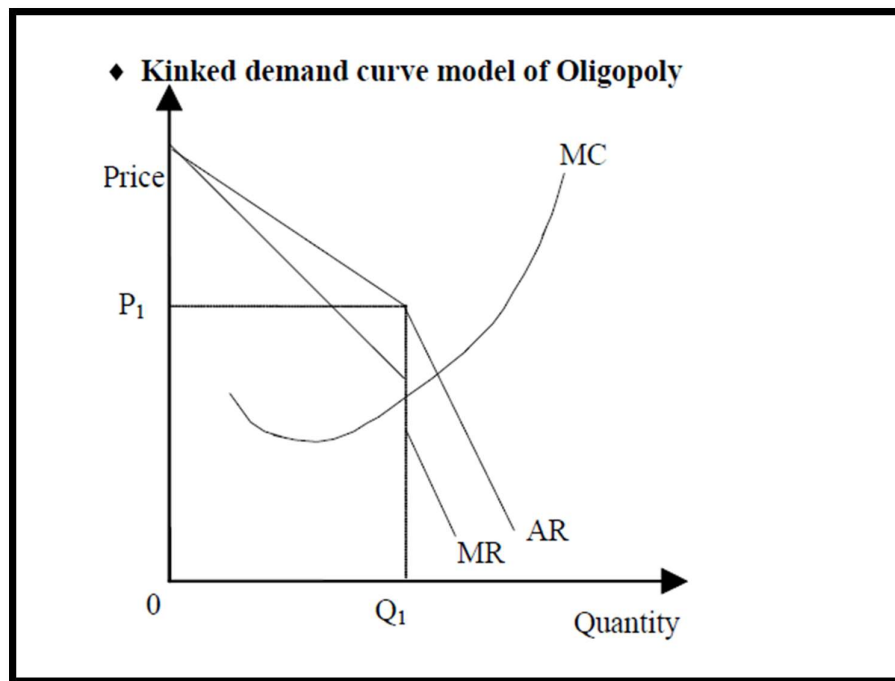
- i) A kinked demand curve occurs when the demand curve is not a straight line but has a different elasticity for higher and lower prices. [0.5]

It suggests that prices are rigid and that firms will face different effects for both increasing price or decreasing price. The kink in the demand curve occurs because rival firms will behave differently to price cuts and price increases [0.5]

If a firm increases the price, then it becomes more expensive than rivals and therefore, consumers will switch to its rivals. For a price rise, there is likely to be a significant fall in demand. Demand is, therefore, price elastic. In this case, firms will lose revenue, the percentage fall in demand is greater than the percentage rise in price. [1]

However, if a firm cut its price the other firms will also cut price to follow the first firm. Hence, the individual firm will only see a small increase in demand. Because there is a 'price war', demand for a firm is price inelastic – there is a smaller percentage rise in demand. If demand is inelastic and price falls, then revenue will fall. [1]
Hence, the firm has no incentive to raise price or to cut price. [3]

ii)



[2]
[5 Marks]

Solution 30:

Quantity	Total Cost	Total Variable Cost	Marginal Cost per unit	Average variable cost	Average total cost
0	50000				
1000	55000	5000	5.00	5	55.00
2000	58000	8000	3.00	4	29.00
3000	59000	9000	1.00	3	19.67
4000	64000	14000	5.00	3.5	16.00
5000	70000	20000	6.00	4	14.00
6000	83000	33000	13.00	5.5	13.83
7000	99000	49000	16.00	7	14.14
8000	122000	72000	23.00	9	15.25
9000	149000	99000	27.00	11	16.56
10000	200000	150000	51.00	15	20.00

- i) The Breakeven price is the price at which the cost of production equals the revenue for a product. (where revenue is price *quantity)

Hence, it would be such that the price is at least able to cover his Average total cost . Looking at the table above, Rs 13.83 (6000 quantities) is the minimum average total cost, and hence, the breakeven price.

Any price below 13.83, Dan would not be able to cover his costs completely.

[2]

- ii) The shut down price is such that Dan earns just enough revenue to cover its total variable costs.

The shut down point for Dan would be where the Average variable cost is the minimum. i.e., Rs 3 (3000 quantities)

Any price below 3, Dan would not be able to cover his variable costs completely.

[2]

- iii) Marginal Cost per unit is calculated as the (Change in Total Cost/ Change in quantity).

The marginal cost at 5000 units is Rs 6. Therefore, Dan should produce 5000 quantities if the price is Rs 7 per pen.

$$\begin{aligned}\text{Net Profit/Loss} &= \text{Total Revenue} - \text{Total Cost} \\ &= 7 * 50000 - 70000 = (35000)\end{aligned}$$

At any quantity greater than 5000 units the marginal cost is going to be greater than the additional revenue.

We can also calculate the profit maximising point by the following table:

Quantity	Total Cost	Total Revenue	Loss
0	50000	0	-50000
1000	55000	7000	-48000
2000	58000	14000	-44000
3000	59000	21000	-38000
4000	64000	28000	-36000
5000	70000	35000	-35000
6000	83000	42000	-41000
7000	99000	49000	-50000
8000	122000	56000	-66000
9000	149000	63000	-86000
10000	200000	70000	-130000

[2]

He shall continue producing in the short run, because he is partially able to contribute to his fixed cost.

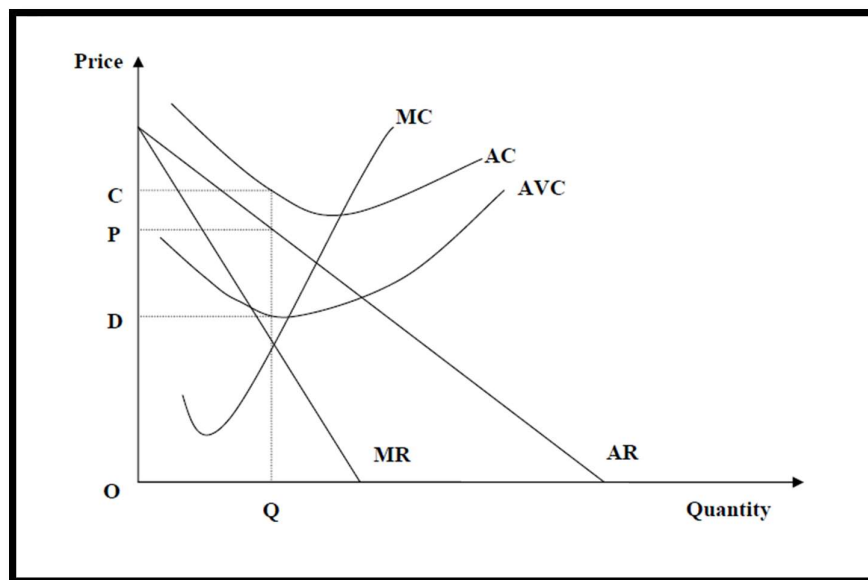
[0.5]

However, in the long run, when none of the cost is fixed, Dan may choose to shut down.

[0.5]

[3]

iv)



[4]

[11 Marks]

Solution 31:

i)

GDP Deflator - GDP deflator is a ratio of nominal GDP to real GDP.

$$\text{GDP Deflator} = \text{Nominal GDP} / \text{Real GDP}$$

Consumer Price Index (CPI) - The CPI is the price of this basket of goods and services relative to the price of the same basket in some base year.

[2]

ii)

Differences between GDP Deflator and CPI:

1. The GDP deflator measures the prices of all goods and services produced, whereas the CPI measures the prices of only the goods and services bought by consumers.

2. The GDP deflator includes only those goods produced domestically. Imported goods are not part of GDP and do not show up in the GDP deflator. Hence, an increase in the price of a good produced in foreign country and sold in domestic country affects the CPI, because the good is bought by consumers, but it does not affect the GDP deflator.
3. The CPI assigns fixed weights to the prices of different goods, whereas the GDP deflator assigns changing weights. In other words, the CPI is computed using a fixed basket of goods wrt base year, whereas the GDP deflator allows the basket of goods to change over time as the composition of GDP changes.

[2]

iii) The key drivers responsible for the slowest GDP growth rate are as follows:

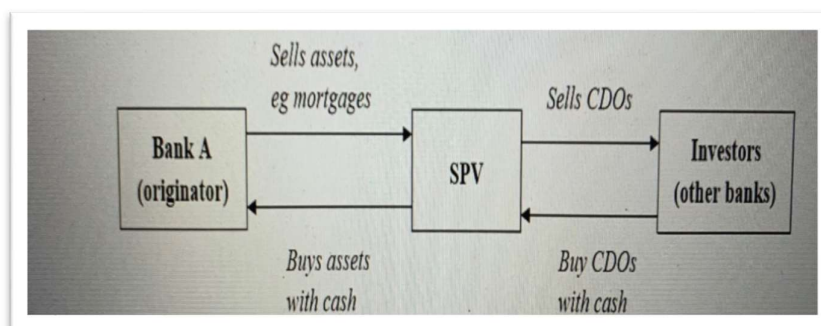
- a. Reduction in growth rate of developed economies
- b. Subdued economic activity in majority of sectors
- c. Reduced growth in manufacturing sector
- d. Investments (gross fixed capital formation) grew at lower rate, reflecting poor sentiment among investors and big companies.
- e. The collapse in private consumption demand in fourth quarter of FY 2017-18.
- f. Government expenditure grew at a faster rate than the economy
- g. There is fall in credit to consumer durables
- h. Sharp fall in domestic consumer expenditure
- i. The sharp slowdown in financial, real estate and professional services sector
- j. Fall in growth of gross value added due to agriculture sector
- k. Fall in food prices in agriculture hurt agriculture income and spending
- l. Failure of financial institutions / bankruptcy of financial institutions
- m. Lower customer confidence in credit systems / financial systems
- n. Sharp fall in liquidity of financial markets (liquidity crunch)

[3]

[7 Marks]

Solution 32:

i) Diagrammatic representation of Securitization arrangement:



[2]

ii) **Role of securitization in financial crisis of 2008:**

- Securitization of subprime mortgages in the US and the UK grew rapidly in the period 2000-2008.
- Bank lending to individuals that was securitized increased from substantially from 1998 to 2008.
- The rise in wholesale funding and securitization provided banks with huge sums for lending but, with higher levels of gearing, greater risk.
- On the other side of the balance sheet, banks' assets became less liquid and riskier as they extended more long-term credit to firms and households, eg 100% mortgages, and to other banks through the sale of mortgage-backed securities.
- Although an individual bank appears to have an increase in its liquidity and uses it to increase its lending, when many banks are doing the same thing,

there is no increase in liquidity in the financial system as a whole. Instead, there is a credit bubble.

- This is known as co-ordination failure
- Towards the end of 2006, there was an increase in the number of defaults and late payments on the sub-prime mortgages. House prices began to fall and banks became concerned about their exposure to these mortgages.
- The resulting uncertainty then led to the “credit crunch” with banks being reluctant to lend, particularly to each other.
- Central banks around the world had to step in to address the immediate liquidity problem, as well as more fundamental concerns.

[4]

iii) **Fiscal policies adopted by Government in response to financial crisis and its aftermath:**

- The fiscal policy yo -yo was caused initially by financial crisis which was triggered by defaults on sub-prime debts.
- Government adopted expansionary monetary and fiscal policies to ensure aggregate demand in economy.
- Government increases spending and reduces taxation and therefore, increased size of budget deficit.
- Within a year or two, the policies were reversed as government became concerned about the size of budget deficit.

[4]

[10 Marks]

Solution 33:

i) Taylor Rule and economic objectives of Central Bank:

- A Taylor Rule takes two economic objectives into account. It gives how much the real interest rate must be raised if :
 - Inflation is above target or
 - Real GDP is above the potential level
- General form of Taylor Rule equation is as follows:

$$r = r^* + b(\pi - \pi^*) + c(Y - Y_p)$$

- Where:
 - r = real interest rate set by Central bank
 - r^* = real interest rate consistent with long-run equilibrium in economy
 - π = current inflation rate
 - π^* = target inflation rate
 - Y = current level of real GDP
 - Y_p = Potential level of Real GDP
 - b and c are positive constants
- From equation, it is implied that the central bank will raise the real interest rate if inflation goes above its target level or if real GDP goes above its potential level with b and c represents weights attached to two economic objectives.

[4]

ii) Definition of dynamic aggregate demand curve (DAD) -

Dynamic aggregate demand curve (DAD) is an aggregate demand curve with inflation on the vertical axis instead of the price level.

With increase in real interest rate when inflation is above the target level, resulting in fall in real GDP and DAD curve gets flattened.

[2]

[6 Marks]

Solution 34:

i) Equilibrium occurs where Aggregate Expenditure $AE =$ National Income Y , therefore:

$$C + I + G + E - M = Y$$

$$(100 + 0.6Y) + 100 + 100 + 40 - (40 + 0.1Y) = Y$$

$$100 + 0.6Y + 240 - 40 - 0.1Y = Y$$

$$340 - 40 + 0.5Y = Y$$

$$300 = 0.5Y$$

$$Y = 600$$

Therefore,

1. Equilibrium output = $Y = 600$ [1.5]
 2. Consumption = $(100 + 0.6 * 600) = 460$ [0.5]
 3. Import = $(40 + 0.1 * 600) = 100$ [0.5]
 4. Trade deficit = (Import – Export) = $100 - 40 = 60$ [0.5]
- [3]**

ii) J curve effect:

This refers to a phenomenon wherein the trade balance of a country worsens following the depreciation of its currency before it improves. [1]

iii) The main drivers for the potential globalization of an industry are:

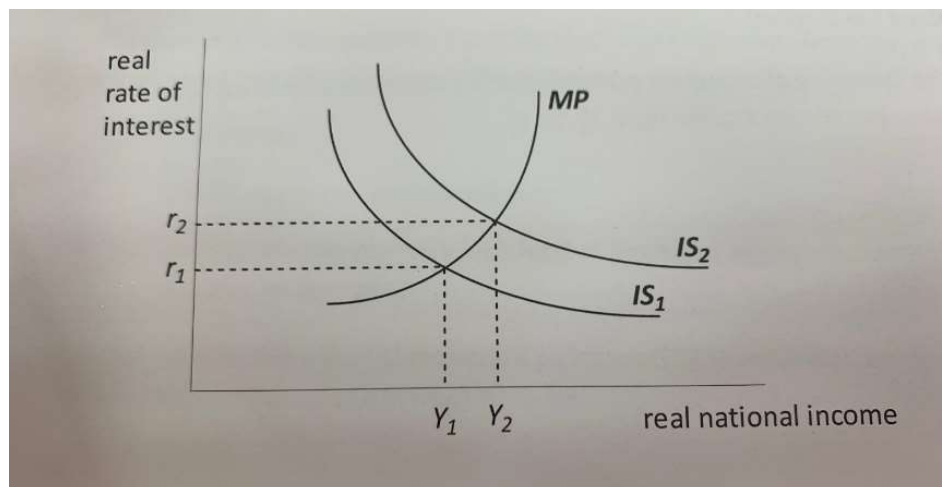
- Market drivers
 - Cost drivers
 - Government drivers
 - Competitive drivers
- [1]

iv) Effect on interest rates and national income using the IS-MP (Investment – savings, Monetary Policy model)

An increase in the demand for exports:

1. An increase in exports raises aggregate demand and real income and shifts the IS curve to the right. (From IS_1 to IS_2)
2. The increase in income leads to an increase in inflation rate, so the central bank raises interest rates (i.e. movement along MP curve) to curb inflation and keep it at target level.

Due to increase in exports, there is an increase in national income from Y_1 to Y_2 and increase in interest rates from r_1 to r_2 .



[2]
[7 Marks]
